

# Student Mental Health Monitoring System

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## SOFTWARE ENGINEERING LAB – EXP-3A

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**Title:** Student Mental Health Monitoring System (SMHMS)

Students across educational institutions increasingly face mental health challenges — stress, anxiety, depression, and burnout — that often remain undetected until they severely affect academic performance and overall well-being.

Traditional support mechanisms depend primarily on reactive, self-initiated counseling visits or periodic surveys, which fail to capture the full spectrum of at-risk students, especially those unwilling to self-report due to stigma.

**Proposed Solution:**

The Student Mental Health Monitoring System (SMHMS) is an intelligent, proactive, multi-modal platform that integrates:

- **Universal Behavioral & Emotional Screening** using validated instruments such as BESS, PSC, and SDQ to identify at-risk students early.
- **Daily Observational Data Collection** — tracking absenteeism, academic performance, sleep duration, screen time, and mood deviations.
- **Machine Learning & NLP-Based Risk Classification** using decision trees, ensemble classifiers, and text analysis.
- **High-Risk Group & Bullying Detection** to identify social and behavioral warning signs.
- **Coordinated Multi-Stakeholder Alerts** notifying counselors, parents, and administrators for timely intervention.

Two new actors are introduced:

- **Parents/Guardians** (consent-based alerts)
- **IT/Data Security Officers** (ethical governance & compliance)

SMHMS aligns with **SDG 3 (Good Health and Well-being)** and **SDG 4 (Quality Education)**.

# 1 ACTORS IN THE UPDATED SYSTEM

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## 1.1 Existing Actors

### 1. Student

Fills screening questionnaires, logs daily mood, views personal health status, and tracks well-being history.

### 2. Counselor / Mental Health Professional

Views at-risk dashboard, accesses detailed student reports, schedules counseling sessions, and provides interventions.

### 3. Admin (Institution Authority)

Manages users, configures screening tools and ML thresholds, monitors institutional mental health trends.

## 1.2 New Actors

### 4. Parent / Guardian (NEW)

Receives critical risk alerts (with student consent), views summarized well-being reports, supports early intervention.

### 5. IT / Data Security Officer (NEW)

Manages access control, ensures FERPA/DPDP compliance, conducts security audits, and enforces ethical data policies.

## 2 UPDATED USE CASE DIAGRAM

### System Overview:

- Total Actors: 5
- Total Use Cases: 20
- Solid Lines represent Actor–Use Case Associations

The updated system expands from a basic monitoring model to a comprehensive five-actor ecosystem, ensuring early detection, secure data governance, and coordinated intervention.

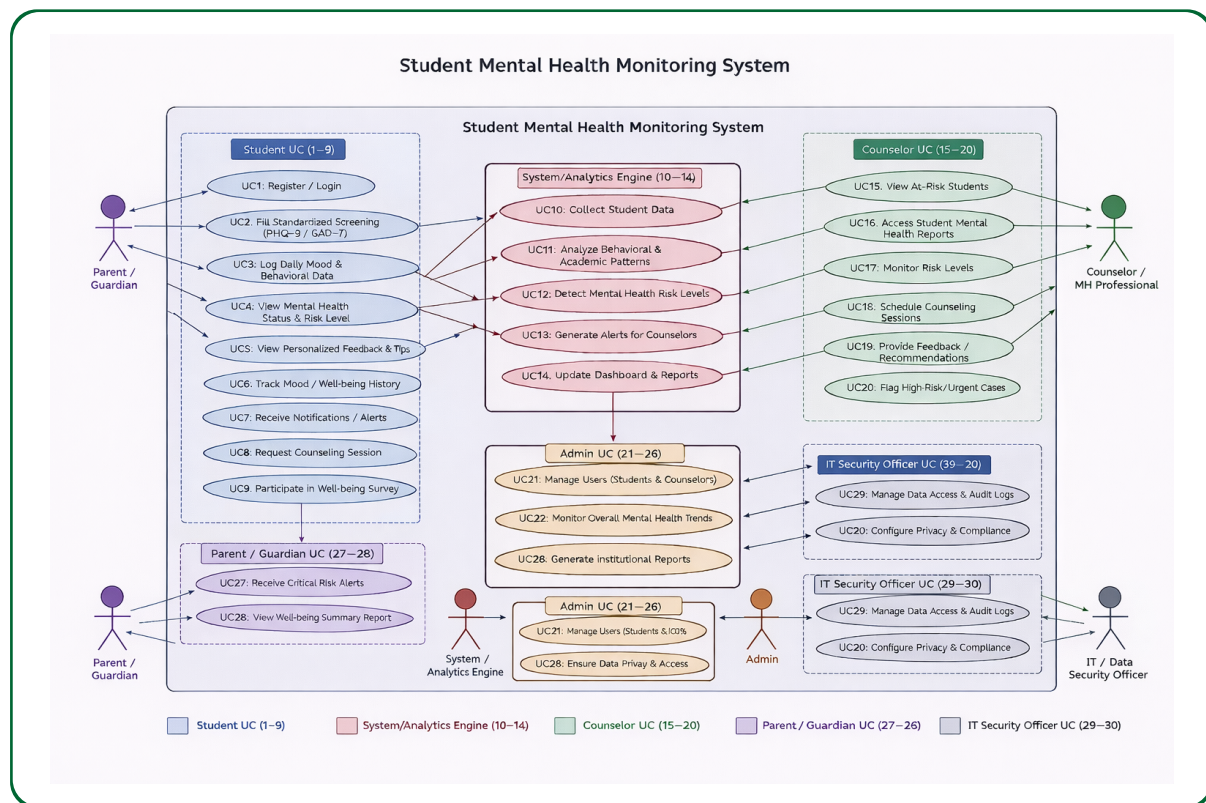


Figure 1: Updated Use Case Diagram – Student Mental Health Monitoring System

### 3 JUSTIFICATION OF NEW ELEMENTS BASED ON LITERATURE SURVEY

|                                                |                 |                                                                                                                                                             |
|------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| primaryColor                                   |                 |                                                                                                                                                             |
| UC2 – Standardized Screening (BESS/S-DQ/PSC)   | Sources 1, 3    | Evidence-based universal screening tools catch students with mild/emerging symptoms that are missed by surveys or academic performance alone.               |
| UC3 – Daily Behavioral Observation Logging     | Source 2        | Daily health observation scores and absenteeism trends are strong early predictors of depression and anxiety before clinical symptoms appear.               |
| UC12 – Flag High-Risk / Bullying Cases         | Source 5        | Bullying significantly raises anxiety and depression risk; structured detection frameworks identify distress that would otherwise be overlooked.            |
| UC14 – Configure Screening & ML Thresholds     | Source 6        | ML-enhanced and redesigned questionnaire instruments yield better early indicators; admin-configurable thresholds allow institutional adaptability.         |
| Actor: Parent / Guardian (UC17, UC18)          | Sources 2, 4, 5 | Coordinated intervention involving caregivers bridges the school-home gap, enabling faster real-world support before a student's condition deteriorates.    |
| Actor: IT / Data Security Officer (UC19, UC20) | All Sources     | Handling sensitive student mental health data mandates dedicated governance roles for ethical compliance — a gap identified across all surveyed literature. |